

Joseph Walton

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Professional Summary

Machine Learning Engineer with over 10 years of experience across machine learning, data engineering and software engineering. Proven track record of building, deploying and improving reliable production ML systems, including GCP and Vertex AI workflows, large-scale data pipelines and NLP models. Strong problem solver experienced in investigating failing AI projects, optimising complex data flows and delivering critical performance and cost improvements.

Technical Skills

- **Programming Languages:** Python, SQL, Java
 - **Machine Learning:** Scikit-learn, TensorFlow, PyTorch, NLP, BERT, recommendation systems
 - **Data Analysis & Visualisation:** Pandas, Polars, NumPy, Apache Spark, Matplotlib, Seaborn, Jupyter
 - **MLOps & Tools:** Vertex AI Pipelines, Azure ML Pipelines, MLflow, Docker, Git
 - **Cloud Platforms:** Google Cloud Platform, Azure, AWS
 - **Databases & Search:** PostgreSQL, MySQL, Elasticsearch
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Professional Experience

IBM

Lead Data Scientist

June 2025 - Present

Contract role with central government.

- Rescued failing AI projects by introducing robust evaluation and observability practices.
- Designed and delivered a production data-processing agent using Azure AI Foundry, Azure Functions and LangGraph, saving case workers two hours per day and approximately £2M annually.
- Rewrote a data-matching algorithm in Polars from Apache Spark, reducing runtime from two hours to ten minutes and cloud costs by approximately £10k per month.
- Optimised human-in-the-loop data ingestion for agent evaluations, saving approximately one hour per evaluation case.

Tech Stack: Python, Polars, Apache Spark, Azure AI Foundry, Azure Functions, LangGraph

Intellectual Property Office

Senior Machine Learning Engineer

November 2021 - April 2025

Developed and operated NLP solutions to improve trade mark application outcomes.

- Implemented an end-to-end MLOps strategy across multiple environments, reducing model deployment time to under one day.
- Architected and implemented ETL pipelines using Polars, replacing an expensive Azure Databricks implementation, reducing compute costs by 50% and enabling zero-downtime deployments.
- Refined production search and recommendation systems using data-driven evaluation, improving relevance metrics by 20%.
- Implemented vector search with Elasticsearch and tuned its HNSW configuration to balance throughput and recall.
- Developed Generative AI applications within a service-oriented architecture using LangChain, Azure OpenAI and Azure Functions.
- Conducted red-team exercises to identify and address weaknesses in Generative AI applications.

Tech Stack: Python, Azure OpenAI, Hugging Face, Azure ML Pipelines, Azure Functions, Elasticsearch, Polars

Incopro

Lead Machine Learning Engineer

February 2020 - November 2021

Led a team delivering production ML solutions for intellectual property protection.

- Designed and implemented an MLOps framework on Google Cloud Platform using Vertex AI Pipelines to orchestrate training, evaluation, approval and deployment workflows.
- Reduced time to market for new models to under one day while supporting client-specific model deployments.
- Built Spark data-processing pipelines on GCP for large-scale text and image model training, batch inference and recommendation workloads.
- Developed and deployed NLP and computer vision models, including a multilingual BERT model for detecting IP infringements across more than 90 languages.
- Implemented a recommendation engine based on text and image similarity.
- Mentored a software engineer transitioning into machine learning.

Tech Stack: Python, Vertex AI Pipelines, TensorFlow, PySpark, Google Cloud Platform, MLflow

Machine Learning Engineer

November 2018 - February 2020

Researched, developed and deployed production ML models for IP protection.

- Engineered a containerised production NLP system for predicting IP infringement, increasing enforced infringements by 10% per analyst across a team of approximately 200.
- Built and deployed an image-similarity model that helped secure a blue-chip client contract worth 10% of annual recurring revenue.

Tech Stack: Python, TensorFlow, FastText, Google Cloud Platform, MLflow

Office for National Statistics

Machine Learning Engineer

September 2017 - November 2018

- Developed ML models and scalable data pipelines to link and process government administrative data for national survey programmes.
- Developed a highly scalable backend for the UK Census 2021 data-collection system.

Tech Stack: Python, PySpark, Cloudera

Barclays

Data Scientist

September 2018 - October 2018

Seconded to Barclays following an ONS x Barclays hackathon win.

- Used payment data to develop regional economic indicators through secure and ethical data analysis.
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Purple Secure Systems

Software Engineer

August 2014 - October 2016

- Developed full-stack, big-data processing systems for government and defence clients.

Tech Stack: Java, Python, PySpark, AngularJS

Education

University of Bath

Master of Science in Applied Mathematics

2016 - 2017

Bachelor of Science in Physics

2010 - 2013

Honors & Awards

Winner of ONS x Barclays Data Hackathon

Used payment data to create more granular and timely estimates of Gross Value Added (GVA).